**Project Design Phase-I**
**Solution Architecture**

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| Date | 24 February 2026 |
| Team ID | 699d5f89c40b5ad0e9ee5981 |
| Project Name | QUOTES RECOMMENDATION CHATBOT USING NLP |
| Maximum Marks | 4 Marks |

## Solution Architecture:

The Quotes Recommendation Chatbot using NLP architecture is designed to process user input, analyse the text using Natural Language Processing techniques, and provide relevant quotes as responses.

The main goals of the solution architecture are:

• To design a system that can understand user queries related to emotions, motivation, or topics.
• To process user input using NLP techniques such as tokenization, sentiment analysis, and keyword extraction.
• To retrieve relevant quotes from a database based on the analysed input.
• To generate a meaningful response and display it through a chatbot interface.
• To ensure the system is scalable, efficient, and easy to maintain.

## Example - Solution Architecture Diagram:

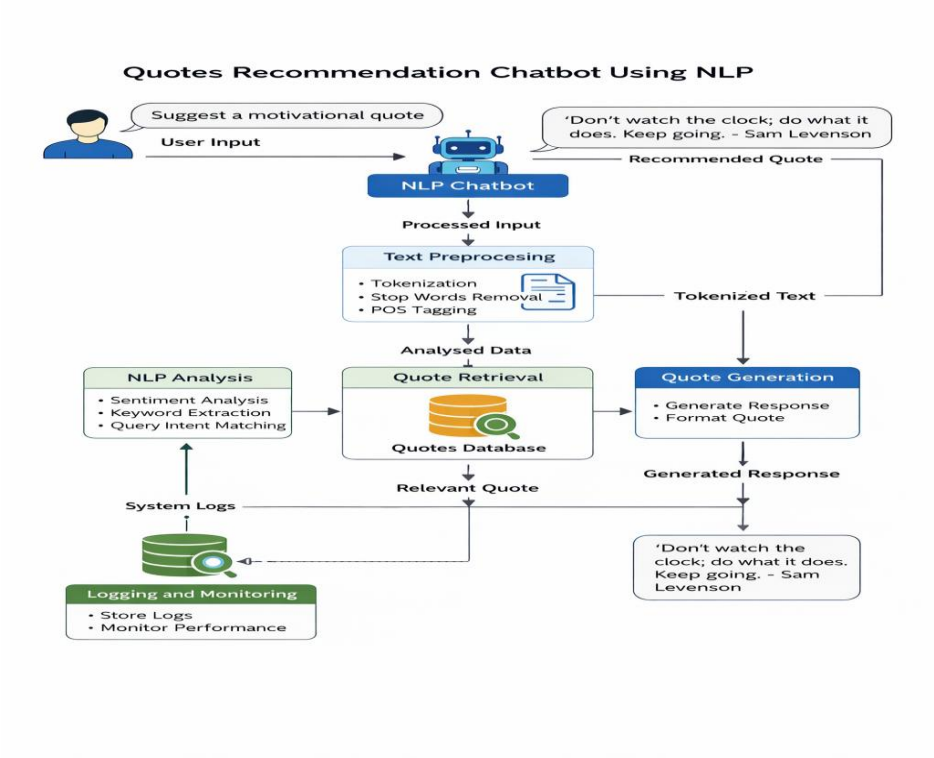

*Figure 1: Architecture and data flow of the voice patient diary sample application*

**Reference:**


1. Swiftorial. *Chatbot Architecture and Components*. Available at: https://www.swiftorial.com/tutorials/project_management/project_integration/building_a_chatbot/chatbot_architecture/
2. Huang, X. *Chatbot: Design and Architecture*. University of Pennsylvania Research Paper. Available at: https://www.cis.upenn.edu/wp-content/uploads/2021/10/Xufei-Huang-thesis.pdf
3. Dragon1. *Chatbot Architecture Framework*. Available at: https://www.dragon1.com/concepts/chatbot